When Schema Lexicons Fail: Diagnosing and Repairing Entity-Layer Collapse in a Production Knowledge Base

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ABSTRACT

In dictionary-driven entity enrichment, the schema lexicon effectively serves as the extraction model. We report a production incident in which a category-word lexicon silently collapsed the entity layer of a 61,137-document scholarly knowledge base: 575 type-level entities ("method", "energy") carrying ∼19.2M links while every job-level monitor stayed green. We diagnose the failure across three layers (extraction mechanism, lexicon provenance, monitoring blindness), repair it with a corpus-attested lexicon built by LLM classification of corpus candidates under presence gating (deployed as a three-source recipe) (3,702 terms), rebuild the entity layer under production load, and derive gold-free monitors led by a concentration diagnostic invariant to uniform rescaling of link counts. A controlled experiment that fixes code and corpus and varies *only the lexicon* reproduces the collapse signature alone (109±5 entities, top-10 concentration 0.513±0.016) and isolates the repair effect (692±27, 0.300±0.008; ten resamples); the rebuilt layer carries 1,871 entities and 25.1M links; distinct concepts per document rise 15.7→37.8 with annotated-document coverage 51.5%→98.5%; a risk-stratified 300-link audit reads 0.84 overall (LLM-assisted; no human gold yet). Matched spans are equally "valid" under old and new dictionaries (0.94 vs. 0.93, model-judged)—the collapse is a vocabulary-and-mass failure invisible to per-link audits—and a composite four-leg monitor rule reaches zero misses and zero false alarms on its own in-sample degradation archive. A three-domain replay on public corpora shows the failure form and the repair travel: batch-schema lexicons reproduce the collapse signature out of domain. Limitations are stated up front: the generic tail survives repair (9% of links sit on the symbol "be"), retrieval-facing gains are not attributable to the lexicon alone, annotation is model-based (Qwen-family readings; cross-family arbitration on 8–29% of rows) with single-author human verification of a 197-term stratified sample (confirm 0.934) — an independent dual-annotator human gold is the declared next step — and alarm bands are post-hoc candidate thresholds.

KEYWORDS

knowledge base construction schema quality terminology extraction entity linking production incident observability

1 INTRODUCTION

Scholarly knowledge platforms increasingly generate subject schemas with LLM assistance: a batch tool drafts entity types with a handful of aliases each, a validator checks structural well-formedness, and the schema is bound to knowledge bases as configuration [15]. When the enrichment pipeline then extracts mentions by *dictionary scan*, the schema's alias list becomes the extraction surface — the lexicon is not descriptive metadata but operative infrastructure.

The failure we report: a 61,137-document nuclear-electronics knowledge base accumulated ∼19.2M entity links onto **575 canonical entities**, every one a bare type word ("method", "parameter", "energy"); a 13,408-document physics library showed the same failure pattern at 2,018 entities. Every operational signal stayed green — jobs done, throughput nominal, link volume reading as growth. The graph had degenerated into a bipartite type–document index that was row-level indistinguishable from a populated knowledge graph.

We contribute three things. **(1) Incident forensics**: the collapse record (§3), the full-chain rebuild under live load (§6), and a corrected incident log demonstrating that dashboard-derived incident narratives themselves require row-level verification. **(2) Single-variable causal isolation of the lexicon**: with code and corpus fixed, swapping the dictionary alone reproduces the collapse signature and its repair across ten disjoint corpus resamples (§4; a frozen-sample parity check ties the replica's matching semantics to the production scanner) and replays out of domain (§8) — the deployed three-source recipe (decisive mechanism: LLM classification of corpus candidates under presence gating) is documented as the treatment under study, not claimed as a new term-extraction method. **(3) Gold-free monitor calibration, led by top-10 link concentration**: candidate thresholds read from 13 production libraries, verified in-sample on a graded-degradation archive, reported with their known false-positive and false-negative classes (§7.1).

2 BACKGROUND AND PRIOR ART

Dictionary- or gazetteer-driven entity recognition is one of the oldest NER traditions [6]; corpus-statistical term recognition goes back to C-value/NC-value [1] — our corpus-candidate source is that program reborn with an LLM classifier, and recent work extracts terminology directly with LLM-plus-retrieval pipelines [5]. The noise side is classical too: gazetteers as both strength and failure surface of dictionary NER [7], UMLS-scale dictionary lookup in MetaMap [8], and distant supervision's false-positive mass [9]; our incident is that literature's production-KB echo at the schema layer — the gazetteer is batch-generated content, and its noise is invisible per-link while fatal in the aggregate. Automatic schema and KG construction has meanwhile advanced to dynamic schema induction over corpora (AutoSchemaKG [2]), incremental schema evolution (DIAL-KG [3]), and joint evaluation of extraction quality, schema adherence, and stability [4]. Our subject is complementary and upstream of a different risk: in *dictionary-scan deployment*, where the schema's alias list is the extraction surface, we show that schema-legality and even per-link mention validity can coexist with asset-level semantic collapse — a failure locus those methods do not measure, and one that our diagnostics (distribution and name monitors, coverage accounting) target directly. We do not claim a better schema-induction or term-extraction *algorithm*; the

Table 1: The collapse, measured, with snapshot provenance ("bckp." = 2026-10-06 cold backup, mid-enrichment; "disc." = discovery-day).

KB	Entities	Links
nuclear (61,137 docs)	**575** (disc.) / 503 (bckp.)	∼**19.2M** (disc.) / 4.81M (bckp.)
physics (13,408 docs)	**2,018**	2.23M (bckp.)
hydro (ref., 78 docs)	1,608	719,515

contribution is the reproducible failure, the separation of local validity from coverage structure, and the repair lifecycle. Knowledge-graph refinement [10] and production data validation [11, 12] address adjacent problems; term-extraction evaluation is annotation-guided and domain-stratified (TermEval/ACTER [13]), and vocabulary drift is measurable even absent failure [14]. What this incident exposes is a specific failure locus: when the dictionary effectively *is* the extraction model, a structurally valid but terminologically hollow schema passed every gate *this pipeline had in place* (schema-shape validation, ingestion checks, job-level monitors). Distribution-statistics monitoring [11] is the right tradition, but the statistic must be aimed at the lexicon's terminological content; none of our gates asked whether a lexicon's entries are domain-specific terms.

Our pipeline extracts mentions in stage K1 by normalized dictionary scan (CJK substring; ASCII word-boundary with optional plural, case-folded); K2 links mentions to canonical entities by normalized form + type; K3 adjudicates typed relations with an LLM; K4 projects to the graph store. The healthy reference case in the same deployment — hydrogeology, 78 documents, a hand-evolved 3,742-term lexicon, 1,608 entities — establishes plausibility, not causation: domain, scale, and provenance all differ. Its ∼22× per-document link-mass edge (∼9,200 vs. 315–410 links/doc) is a length artifact — textbook-scale ∼1,913 chunks/doc vs. KB1's 99 — with comparable per-chunk link rates (4.8 vs. 4.1).

3 THE COLLAPSE, FORENSICALLY

Observation (Table 1). During an unrelated storage investigation (2026-10-08), a routine entity-quality sample returned type words where domain concepts were expected.

Mechanism. When the alias set for type *method* contains common words, every occurrence in 7.2M chunks becomes a mention, and K2 folds them onto one entity per (form, type). The mechanism executed faithfully on degenerate input; the dictionaries were the schemas — the batch generator produced a category taxonomy where a terminology was needed [16].

Timeline (compressed): 10-08 discovery → root cause, lexicon tool built/run/committed, teardown + 766,501-event re-enqueue; 10-08 night: silent mis-attribution window (below); 10-09 drain completes, statistics recomputed, two smaller libraries with the same failure pattern treated; 10-09/10: retrospective audits (scoped refresh, channel ablation, link audit, fleet-wide concentration) and manuscript. The compression is real; the evaluation is retrospective, not pre-registered.

Why nothing fired. Job-level observability was all green; link volume read as success. The observables that expose the collapse are the concentration of links on the top entities (collapsed: 0.419, measured from the pre-incident backup, versus 0.18 for the reference case) and the name distribution of those entities (pure type vocabulary). Entity cardinality per document — a factor of 2,192 below the reference case (0.0094 vs 20.6) — corroborates, but is gameable in both directions, so it is the secondary signal.

4 IS THE LEXICON THE CAUSE? ISOLATING THE VARIABLE

The repair bundled lexicon, teardown, and fleet changes; this experiment isolates the lexicon alone *before* we present the repair method: a semantic replica of the K1/K2 scan-and-link semantics with *only the dictionary swapped* — v2.0 as committed at library creation (182 scan terms), v2.5 mid-evolution (2,932), v2.1 the repair (3,733 scan terms;[1]; construction in §5), plus two v2.1 governance variants. The replica copies the scanner's matching and folding rules from the same source reading (CJK substring; ASCII word-boundary with optional plural, case-folded; fold key = normalized form + type); it is not instrumented against the live service path, so absolute levels may drift — the experiment's force is the within-frame arm contrast. A frozen-sample parity check bounds that drift: on 200 frozen KB1 chunks scanned with the *live* production dictionary (3,824 entries), every one of the 913 production mention spans is reproduced by the replica (span-level *superset*, not output equivalence: the per-term replica additionally reports 263 overlapping or nested occurrences that the production merged longest-first alternation suppresses by design, and a normalized-form bookkeeping rule differs for aliases that themselves end in "s"). Boundary, case, plural, and CJK semantics are identical; the replica's link counts are an inclusive upper bound within that frame. Because nesting prevalence can differ per lexicon, we re-ran all cross-domain arms under production-style overlap suppression (longest-match, non-overlapping): links fall 10–40% per arm but *every ordering conclusion is unchanged* (schema CS C_{10} 0.859 with identical entities; the size-matched arm C_{10} 0.323 vs. three-source 0.366; frequency still floods at 23+ concepts/doc).

Design. Ten seeded, uniformly random resamples of 2,000 chunks each from the library's 5.83M visible chunks — independent draws from a *single* corpus, i.e. a sampling-stability check within one library, not cross-domain evidence — cross-sample disjoint, deduplicated on redraw (an earlier single run had sampled an ordered prefix, which we replace rather than defend). Fixed code, fixed corpus, fixed scanner; the only variable is the dictionary. The sampling unit is the chunk; in this corpus chunks map to documents almost one-to-one (sample 0: 2,000 chunks → 1,965 distinct documents, 1.02 chunks/doc), so "matched documents" below means matched chunks and no document is sampled twice. All metrics are deterministic; Table 2 reports mean±sd across the ten resamples. Concepts-per-document is averaged over *matched* documents only (≥1 scan hit under that arm's dictionary: mean

[1]Three v2.1 counts appear in this paper and refer to different objects: 3,702 canonical schema terms (the committed lexicon), 3,733 scan terms after alias dedup (the replica frame), and 3,824 entries in the production composite dictionary (schema lexicon ∪ operational seed overlay, §8's parity check).

Table 2: Controlled lexicon-only comparison across ten disjoint corpus resamples (mean±sd). Conc/doc = concepts per *matched* document; arms' denominators differ (977 vs. 1,579 matched docs per sample); on the full 2,000-document denominator the arms read 0.90 vs. 3.07 (×3.4) — matched-subset conditioning favors the weak arm. sd is within-corpus resample variance only.

Arm	Links	Ents	C_{10}	Conc/doc
v2.0 collapse-era	2,969 ± 140	109 ± 5	0.513 ± .016	1.85 ± .05
v2.5 mid	4,823 ± 147	421 ± 25	0.322 ± .009	2.51 ± .05
v2.1 repair	10,042 ± 251	692 ± 27	0.300 ± .008	**3.89 ± .07**
+case-sens. short	8,758 ± 228	665 ± 28	0.265 ± .012	3.54 ± .07
+stop-word block	9,226 ± 251	691 ± 27	**0.252 ± .011**	3.67 ± .07

977 matched documents per sample under v2.0, 1,579 under v2.1), so the arms' denominators are different sets.

Finding 1 — the collapse signature is reproduced by the lexicon alone. Under v2.0, the same code on the same corpus yields 109±5 entities at concentration 0.513±0.016 with 1.85±0.05 concepts per *matched* document: the collapse needs no pipeline bug, no data loss, no misconfiguration — only a category-word dictionary.

Finding 2 — the repair effect is isolated and monotone. v2.1 multiplies link-bearing entities ×6.3 and concepts per matched document ×2.1 (matched docs per sample 977→1,579); generic-head mass falls 0.146→0.058. The production collapsed concentration (C_{10}=0.419) sits between the v2.0 (0.513) and v2.5 (0.322) counterfactuals, consistent with mid-enrichment dictionary evolution. These are *structural* signatures: entity count, concentration, and concepts per document do not by themselves measure terminology recall or sense fit — the only semantic reading in this paper is the link audit of §7.

Finding 3 — the be class is treatable at known cost. Case-sensitive short symbols remove the largest entity at the price of match volume (links −12.8%; C_{10} →0.265); a stop-word block is gentler (links −8.1%, C_{10} 0.252). The deltas measure matching *output*, not true recall: the removed mass mixes errors with correct terms, and the 0.580 pre-intervention judged-valid rate on this class (§7) motivates the treatment — the removed-vs-retained verification below prices exactly this cut. *Provenance*: decomposing v2.1's link mass by which committed lexicon first contained each term — v2.0-inherited 22.0%, mid-evolution 16.0%, **three-source-new 61.9%** (384 of 686 entities) — is a decomposition, not a per-source causal ablation; the public-domain cached-component ablation (§8) supplies the single-source analogue, while independently regenerating single-source lexicons for the *production* library remains future work. And the arms vary the dictionary as a whole (~20× size gap), so term count, selection quality, and per-source method are not separately identified — the equal-size frequency control and the blind old/new matched-context judging below are the two controls that close this gap on the same frozen samples.

Budget sweep, closing the size confound. A natural concern is whether the isolated comparison (182-term collapsed vs. 3,733-term repair dictionary) confounds size with selection. We re-ran the frozen KB1 frame at fixed budgets $K \in \{200, 500, 1{,}000, 2{,}000, 3{,}733\}$ with three same-K sources: the v2.1 pipeline in construction order (prefix), frequency-top-K, and C-value-top-K (candidates ranked on sample-0 text; frame overlap disclosed). Coverage under the dual-annotated reference grows monotonically with v2.1's own budget (R 0.004→0.043 from K=200 to 3,733) while concentration falls 0.73→0.30 — the repair's value accrues with budget but its structure at every budget is distinct: frequency-top recruits 4–5× the entities (2,036 at K=2,000) for nominally higher R that rides on mass its own blind validity disqualifies (0.45 at the equal-size control), and C-value-top saturates at 554 entities with R stuck at 0.012 — statistical salience without attestation stalls on this corpus. Re-running the sweep with the baselines re-typed by the unified classifier leaves R identical at every budget (entity counts within 1.5%)—typing- invariant. Size explains the counts; it does not explain the validity or the distribution.

Verification: semantics, governance, monitors. Three follow-ups on the same frozen samples close the declared gaps. (i) *Blind term-mention judging* (entity-uniform; judge Qwen3.6-35B, temp 0, disclosed family): old and new dictionaries are *locally* indistinguishable (0.940 [0.890, 0.968] vs. 0.927 [0.873, 0.959] of matched spans are genuine term mentions) — the collapse is not per-link falsehood but a vocabulary-and-mass failure, which is why per-link auditing alone could not have caught it; an *equal-size frequency-only control* (3,733 terms, top corpus frequency, no selection) reads 0.450 [0.356, 0.548] — size does not buy validity; selection does. (ii) *Governance verified*: short-symbol hits removed by case-sensitivity are 0.033 [0.009, 0.114] valid vs. 0.700 [0.575, 0.801] for retained ones — the −12.8% link cut removes almost purely error mass (retained still carries 30%). (iii) *Graded degradation* (replacement axis: drop m% of v2.1, keep all category words): the C_{10} band fires at ≥75% substitution (0.371) and the full collapse reproduces (0.513, bitwise-identical arm); random shrinkage fires count monitors first (links −50% at 50% shrinkage) while C_{10} stays flat to 50%; moderate substitution (10–50%) is a blind zone for both — no single gold-free monitor covers it; the stack deploys as a set.

Short-symbol ambiguity: four linking policies. On 150 short-symbol mentions sampled from the frozen replay chunks (60 on the be class; each mention judged once, blind, by judge A and the cross-family judge, 12 disagreements), four policies on the same dictionary: linking everything (current behavior) reads precision **0.24**; exact-case gating (tested here, not yet deployed) 0.725 at true-symbol recall 0.806, rejecting 73% (rejection precision 0.936); adding context evidence (long-term neighbors) recalls slightly more (0.833) at lower precision (0.652); context plus reverse-NIL trades recall (0.583) for precision (0.724). On the be class specifically, *zero* of 60 sampled mentions are genuine element references — the exact-case gate removes pure error there. Context gating and case gating exchange precision for recall rather than dominating each other; the recovered mass lives in the non-be short symbols.

Why 692 entities here vs 1,871 in production (§7)? The isolated frame counts only entities induced by scanning 2,000 chunks (0.034% of the library) under replica folding: rare terms need corpus mass to be attested, and the production layer additionally carries entities from the full corpus and historical merging. The

causal claim rests on within-frame comparisons, never on this cross-frame count.

5 CORPUS-ATTESTED LEXICON CONSTRUCTION

The repair lexicon whose causal role §4 isolated is built from three sources, all gated by corpus presence (a term the corpus never attests is not admitted — presence is necessary, not sufficient, for termhood): **seed transplant** (2,284 terms from a healthy near-domain library); **corpus-candidate classification** (top 8,000 unigram/bigram n-grams by chunk frequency over a 3,000-chunk-per-KB sample, frequency ≥5 after boilerplate and stop-word filtering; an LLM classifies each into a schema type or rejects it); **per-type expansion** (≤60 per type). Catch-all types accept no classified noise; cross-type conflicts resolve by specificity. Classifier: Qwen3.6-35B-A3B (AWQ 4-bit, community build; served under a historical alias), temperature 0.2, single idempotent run. Output: nuclear v2.1 = 3,702 canonical terms; physics v1.1 = 1,279.

6 REBUILD AT SCALE — AND A CORRECTED INCIDENT LOG

Per knowledge base: delete graph vertices with edges, the four relational tables in bounded chunks, and the search-side projection; re-enqueue all documents through four stages. The printed "766,501 jobs" counts *events*, not distinct doc-stages (74,545×4 = 298,180): the surplus is the repair window's re-enqueue waves (silent-empty reset 75,831; zero-effect relations/graph resets ≈149k; watchdog-deletion recovery ≈122k; small libraries 5,772; ≈115k balanced re-dispatch remainder — teardowns reset the attempts counters, so only the logged resets are exact; the archival long version carries the full reconciliation table).

Four pipeline pathologies surfaced and were fixed with measured effect — an unbounded projection join that saturated the database (25+ min → seconds once scoped), two link-insertion deadlock classes (deadlocks 1213 → 0), and an $O(N^2)$ alias check the richer lexicon itself amplified (fleet link throughput 8/min → 1,100/min, a window-level aggregate). The archival version carries the full per-pathology accounting.

The fifth pathology, revisited. Our log recorded a parameter-order bug in a single-statement link writer whose rows `INSERT IGNORE` "silently swallowed" for seven hours. A minimal reproduction on the production engine (MySQL 8.4, `STRICT_TRANS_TABLES`, faithful DDL) sharpens the mechanism: the NULL from `UNHEX('alloy')` is **replaced by the implicit default 0x00×16 and the row is inserted** — into a ghost bucket invisible to every per-KB query — and a later correct re-insert for the same (mention, entity) pair is **silently skipped by the unique key**: the orphan poisons the repair unless deleted first (production carries zero 0x00 rows today, so cleanup happened). The duration claim also fails row-level forensics: the registry's own `created_at` histogram (Figure 1) shows KB1 inserting at 1.0–2.3M rows/hour *through* the nominal silent window and KB2's entire link layer landing in one hour. We keep the log entry and its correction side by side: incident narratives written from job dashboards are themselves unverified claims.

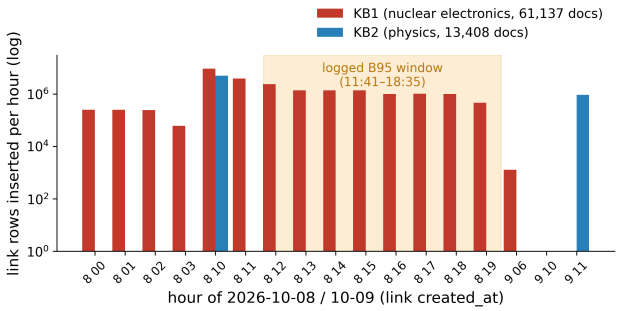


Figure 1: Rebuild telemetry from the link registry's own timestamps. The shaded band is the *logged* silent window; inserts continue through it — the timestamp correction of §6.

Table 3: Entity layer before/after, with snapshot provenance.

Metric (KB1)	Collapsed	Repaired
canonical entities	575 / 503 (10-06)	**1,871**
entity links	~19.2M / 4.81M	**25.1M**
typed relations	~0	**44,043**
top-10 concentration	0.419	**0.345**
concepts/doc (all-doc)	15.7 (51.5% covered)	**37.8** (98.5% covered)
KB2: concepts/doc	29.5	**61.4**

7 EVALUATION

Entity layer, same-snapshot framing. The before/after rows mix two pre-repair snapshots; the like-for-like comparison is the 10-06 cold backup vs. post-repair (503→1,871 entities, 4.81M→25.1M links); the discovery-day figures (575, ~19.2M) are reported for the incident record, as mid-enrichment evolution continued after the backup. Concepts-per-document replaces the raw entity/document ratio: its numerator scales with documents and the coverage column exposes the collapse's most user-visible symptom — half the library had *no* annotations. Absolute link counts *rise* after repair (KB1 19.2M→25.1M; KB2 2.23M→5.0M): real terms match more often. Count was never the quality signal; distribution and names are.

Name quality (Figure 2). The repaired head is still generic: `be` (2.26M links, 9.0%), `alpha`, `gamma`, `beta`, `method`. The `be` case is a normalization failure — element symbol and auxiliary verb merge under case-folded matching — and §4 prices its treatment. The repair moved the problem from "no terminology at all" to "terminology plus a generic tail."

Link audit: 300 risk-stratified links. Entities drawn uniformly within four strata (top-10 by frequency; medium, ranks 11–100; tail, ranks 101+; short symbols, canonical names of ≤4 letters), then one link uniformly per entity — seeded and replayable. Span integrity 300/300. The judged-valid rate is **0.837** for this sample (Wilson 95% CI [0.791, 0.874]; entity-cluster bootstrap over the 187 clusters, 10,000 resamples: [0.782, 0.887]); by stratum: top-10 **0.920** [0.812, 0.968], medium 0.850 [0.767, 0.907], tail 0.910 [0.838, 0.952], short-symbol **0.580** [0.442, 0.706]. Three scope

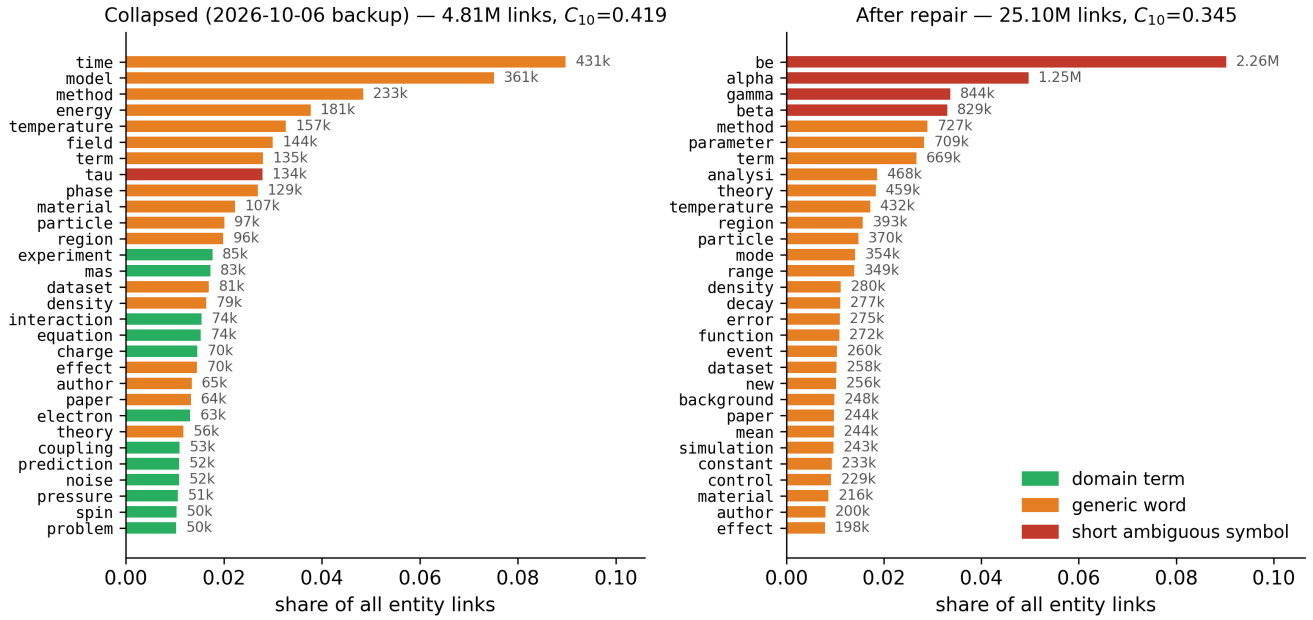


Figure 2: Top-30 entities by link share, before (cold backup; C_{10}=0.419) and after repair (C_{10}=0.345). Categories follow a fixed three-way rubric (domain term / generic academic word / short ambiguous symbol), assigned by the authors per entity; the rubric and per-entity assignments ship with the artifact.

notes. First, the 50/100/100/50 quotas target risk, not production link frequencies: 0.837 is a stratum-mixed rate and must not be read as an unweighted production-wide link precision. Second, the 300 links come from 187 distinct entities (the head stratum's 50 links from just six; the short and head strata share "term"), so the intervals treat links, not concepts, as exchangeable — the CI is neither an interval for the 25.1M production links nor for 187 independent concepts. Third, the design's purpose is to localize high-risk failure — which it does: the short-ambiguous-symbol class that Figure 2 colors red and §4 prices is where validity collapses — a more informative finding than any overall figure. A second Qwen-series model re-adjudicated the earlier uniform sample: on the 55-link parseable subset agreement is 96.4% (domain 14/14) — a cross-model check, not family-independent confirmation (Figure 3), and unparseable responses may be non-random. Human adjudication remains the decisive future control.

Retrieval: honest attribution. Scoped coverage rose +6.6/+7.3 points over the repair window, but three controls bound the attribution: (a) removing the direct entity channels under the final configuration leaves @10 any-hit and document coverage unchanged on both scoped sets (MRR +0.0006, CI [0, 0.0018]; cov@20 −1.0pt on one set) while cutting p50 latency 1.46 s→0.52 s — and a forced-trigger check on the natural-phrasing set shows the channels are alive (54/99 fired; killing them costs 20 losses / 0 wins), so the ablation bounds these channels' marginal contribution in this configuration, not the semantic assets' value at large; (b) alias expansion,

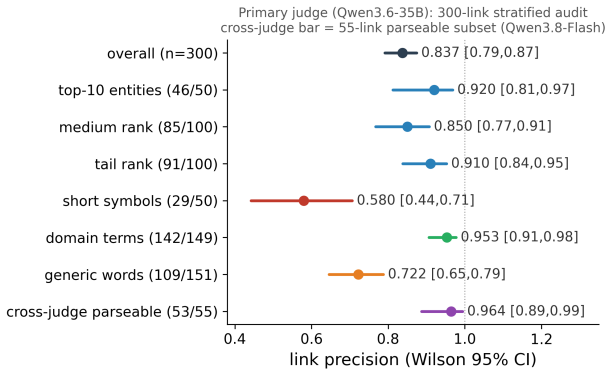


Figure 3: Judged-valid rates with Wilson 95% CIs: the risk-stratified 300-link sample (stratum-mixed, 187 entities; primary judge); the cross-judge bar applies to the 55-link parseable subset of the earlier uniform sample.

the one entity-adjacent mechanism, is +1.5pt (ns) on the benchmarks at 2× latency; (c) two same-configuration runs differ on exactly one query (0.5pt noise floor). The trajectory is system-level; the entity layer's retrieval-facing value runs through asset surfaces (documented in the companion system manuscript [17]), not fusion votes.

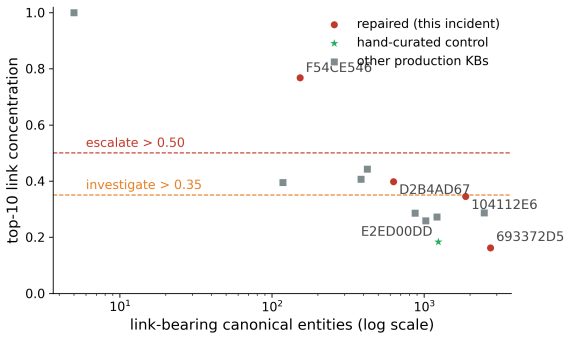


Figure 4: Top-10 link concentration vs. link-bearing entities across 13 production libraries; red = rebuilt in this incident, green star = hand-curated reference. Bands are candidate thresholds read post-hoc from this deployment.

Table 4: The candidate calibration behind Figure 4, summarized (the collapsed row is the pre-repair backup, not a live point).

Group	Libs	Entities	C_{10} range
Collapsed (backup)	1	503	0.419
Hand-curated control	1	1,239	0.184
Other prod., ≥500 ents	4	874–2,482	0.259–0.287
Rebuilt here, large	2	1,871 / 2,726	0.162–0.345
Rebuilt here, small	2	153–629	0.398–0.768
Other prod., <500 ents	4	5–424	0.395–1.000

Snapshot 2026-10-12; the rebuild-completion count for the 2,726-entity library is 2,730 (four-entity post-rebuild drift), cf. §6.

7.1 Monitors and Discussion

Four gold-free monitors, in the spirit of production data validation [11]: **(a) top-10 link concentration** (primary): invariant to uniform rescaling of counts, but bounded below by $\min(10, N)/N$ — trivially 1 for a library with ≤10 entities — and sensitive to topic concentration as well as to degradation. Across the 13 deployment libraries the metric spans 0.16–0.77 among libraries with ≥100 entities (the smallest, five entities, reads 1.000 — the floor bound, not degradation; Table 4); healthy/repaired large libraries run 0.16–0.19, the rebuilt nuclear library 0.345, the two small treated libraries *currently* 0.398 and 0.768 (reported as measured). KB2's repaired 0.162 sitting *below* the control's 0.184 is not outperformance: its thin repaired lexicon (1,279 terms, a declared floor) reshapes head share for reasons orthogonal to term quality. The collapsed physics backup reads 0.290 — *below* the 0.35 band — so a concentration threshold alone does not catch every collapse; the name-audit and coverage legs carry the rest. The bands (>0.35 investigate, >0.5 escalate) are *empirical, post-hoc* candidate thresholds read from these 13 libraries — triage triggers for human inspection, not classifiers, and not out-of-sample validated. Two detection targets must be kept distinct: *collapse detection* (has the entity layer degenerated?) versus *lexicon-quality warning* (is the distribution contaminated with generic vocabulary?). For collapse detection a genuinely concentrated healthy library is a C_{10} false positive; as a lexicon-quality warning the same trigger is correct when the tail is in fact generic. The composite's zero-false-alarm claim below is scoped to collapse detection. Known blind spots (per target): *false positives* (collapse target) — healthy libraries whose terminology is genuinely concentrated, e.g. the two small treated libraries at 0.398/0.768 (real terms, small corpora) and the public-domain CS three-source arm at 0.404 (§8); *false negatives* — replacement-type degradation below ~75% substitution, where C_{10} moves less than one sd (graded degradation), and any collapse whose head happens to be diversified. Validation need not re-collapse production: the graded-degradation arm above injects lexicon decay offline into a frozen slice and reports fire bands and size sensitivity (the experiment has no time axis, so we make no detection-latency claim). A leave-one-library-out check on the 14 rows (thresholds re-frozen at the q90/q97 of the remaining healthy libraries each fold) yields 12 correct quiets, the same known false-positive class (the genuinely concentrated small library), and the same known false-negative class (the collapsed backup reads below the LOO band — count monitors and the name audit carry it): the bands are self-consistent across libraries but remain deployment-derived; unseen-deployment validation stays future work. *Composite rule, evaluated.* Because single-leg thresholds misfire in both directions, the deployable unit is a four-leg disjunction (fire on any leg): concentration (C_{10}>0.35, small libraries exempt), link-count drop (>50% vs. healthy baseline), generic-head share (>0.10), and entity-cardinality drop (>50%). On the frozen graded-degradation archive this composite fires on every should-alarm state (collapse, ≥75% replacement, ≥50% shrinkage) with *zero* misses and zero false alarms on the healthy baseline and mild degradations — reported as in-sample diagnostic verification (thresholds and classes derive from the same libraries; frozen-threshold unseen-library validation is open); the 10–50% replacement band remains undetected by every leg (at 50% the generic-head leg sits at its threshold, 0.096 vs. 0.10) — the honest structural floor of aggregate-statistics monitoring, and the reason the name-audit leg exists. *Threshold sensitivity*: within $\tau \in [0.08, 0.12]$ the generic-head leg's verdicts are unchanged (healthiest archive reading 0.073, a margin ~7× its resample sd); the 75%-replacement state clears three legs at once (C_{10} 0.371, generic head 0.104, links −58.6%), while the 50% state sits four thousandths under τ=0.10 — reading the band down to 0.09 catches it with no new false alarm on this archive. *Cold start*: the two count legs need a healthy history baseline a new library lacks; until a first full enrichment pass seeds it, they are inert, and protection rests on the two history-free legs—concentration and the name audit—exactly the legs that exposed this incident. **(b) entity cardinality ratio** (coarse secondary, gameable both ways). **(c) top-entity name audit** (five-minute quarterly control). **(d) target-table growth reconciliation** — the B95 ghost-bucket mechanism makes this mandatory wherever a bulk writer is guarded by `INSERT IGNORE`. The small-library treatment is an illustrative within-deployment case (the candidate bands were read after observing these 13 libraries — not out-of-sample validation): cardinality normalized (870→629, 209→153) while concentration still flags both — the two monitors separate "cardinality repaired" from "distribution still skewed."

The incident is not an argument for LLM-everywhere: if extraction is a scan, the dictionary serves as the model and deserves a trained model's evaluation discipline. Admission criteria (corpus presence) are necessary but not sufficient; *ranking* criteria (IDF-style contrast, bilingual consistency, length/type priors) are the open problem — the generic tail is the standing evidence.

8 DOES IT GENERALIZE? THREE PUBLIC DOMAINS

The incident corpus is nuclear-science literature; a natural objection is that the failure and the repair are local accidents. We therefore replay the lexicon-only design on three public domains — arXiv cs.CL (10,000 abstracts), PubMed biomedicine (9,988), and arXiv cond-mat.mtrl-sci (10,000) — with seeded splits per domain: 3,000 abstracts for lexicon construction only, 100 frozen test abstracts for silver annotation, and the remainder replayed as ten disjoint ∼690-document samples for structural metrics (mean across samples; sd archived). *Mass* throughout denotes two reported quantities — lexicon size (|L|) and per-document annotation volume (concepts/doc) — plus matched-document coverage where stated; we report all three per arm and predefine no scalar utility over them, so no "best" is asserted. Five arms share one scanner; only the lexicon varies: *batch schema* (the production-failure form: each domain's *own* current repository schema lexicon, i.e. the batch tool's committed output for that domain — category-shaped alias sets, in CS largely Chinese or uppercase-acronym forms), *frequency* top-K and *C-value* top-K (each size-matched to ours), *LLM direct* (per-batch extraction without corpus validation), and the *three-source* recipe. One adaptation is disclosed: source 1 transplants seeds from a healthy near-domain library, which does not exist for these public domains — we substitute LLM knowledge seeds under the same corpus-presence validation; sources 2–3 (corpus-candidate classification, per-type expansion with presence validation) are unchanged. The replay harness reuses the companion system's tooling [17].

Model-annotation protocol with human verification. The annotation layer is model-based; human input enters as verification of a stratified sample (below), not as from-scratch labels. Silver reference lists come from a strong annotator LLM (judge A, Qwen3.6-35B) enumerating every domain term in each test abstract; arm quality is then judged blind on stratified samples (head/medium/ tail/short-symbol entities) by judge A and a second Qwen-series model (judge B), with disagreements and missing judgments arbitrated by a third *cross-family* judge (DeepSeek; 8–29% of rows per cell, Wilson intervals below). Two instrument limits are disclosed rather than hidden. First, enumeration is the weak link: on 30-document subsets re-annotated by the cross-family model, term-list Jaccard with judge A is 0.41 (CS), 0.55 (biomedicine), 0.52 (materials) — so silver P/R are indicative levels, not ground truth, and we read contrasts between arms (which share the instrument) rather than absolute magnitudes. Second, judge B cannot enumerate at all (the gateway forces chain-of-thought past read timeouts) and is used only for short blind judgments; and the LLM-direct arm's silver-P is same-family inflated, since judge A's family also built that arm.

Instrument sensitivity, tested on the shared subset. Because a shared instrument does not guarantee arm-wise bias cancellation, we recomputed all five arms' occurrence-level P/R on the exact 30-document A∩C subset under *both* references (matching rule: lower-cased surface with naive plural stripping, applied symmetrically). The recall ordering of the five arms is **identical under both judges in all three domains**; the F1 ordering is identical in biomedicine and materials, while in CS only the middle band permutes (frequency/C-value/three-source swap ranks 2–4; first and last are unchanged). Three-source's precision *rises* under the cross-family reference in all domains (CS 0.218→0.312, biomedicine 0.195→0.287, materials 0.246→0.260) while LLM-direct's falls (CS 0.653→0.544), consistent with the disclosed family bias and mildly favoring three-source under the harder reference. We therefore rest cross-domain conclusions on (i) structural metrics (no instrument), (ii) recall ordering and (iii) arbitrated blind validity, and treat mid-band F1 orderings and absolute P/R levels as instrument-conditional. Judged-valid rates carry Wilson 95% CIs — e.g. biomedicine batch-schema 0.906 [0.810, 0.956] vs. three-source 0.850 [0.756, 0.912]: overlapping — and the arbitration fraction per cell is 0.08–0.29. *A dual-annotated reference, and what survives it.* To strengthen the instrument we finally cross-annotated 200 complete units — 80 frozen KB1 chunks plus 40 test documents per public domain — with two independent model annotators reading full text (not approving candidates): span-level entries with termness, specificity, type, and identity-ambiguity flags. P1 = deepseek-flash, P2 = deepseek-v4-pro (same vendor, different models — overlap disclosed; a gateway thinking-mode model was attempted first and is unusable for full-document annotation: forced deliberation exceeds any practical token budget), disagreements arbitrated by the cross-family Qwen judge. Agreement is materially stronger than single-model enumeration: domain-set Jaccard **0.59–0.66** across the four scopes, termness agreement on overlap **94.5–98.8%** (vs. 0.41–0.55 for enumeration pairs). Under this reference: the batch-schema arms recall 0–1% (the collapse signature under the strongest instrument yet); the production lexicons' first recall estimate on KB1 reads v2.0 1.0% → v2.1 4.3% of adjudicated distinct concepts (4.3×; the reference is deliberately exhaustive, so absolute levels are floors, not ceilings); the selection arms' recall deficit replicates (learned/cover 0.7–3.2%); and the frequency arm's apparent recall rides on mass its own blind validity (0.29–0.40) already disqualifies. *Human verification*: one author independently confirmed a seeded, stratified sample of 197 adjudicated terms over 20 units (5 per scope; sheet, protocol, and per-row verdicts in the artifact) — confirm rate **0.934** [0.890, 0.961]; by scope 0.833 (KB1) / 0.967 / 0.925 / 0.950; two-model consensus items verify at 0.949 vs. 0.853–0.960 for arbitrated additions. This bounds the reference's precision from the model side; it does not measure omissions (verification, not from-scratch re-annotation) and is single-author — an independent two-annotator gold remains future work. The protocol, adjudications, and per-unit records ship with the artifact. Entity-cluster bootstrap CIs for the three-source validity rates (10,000 resamples) are [0.747, 0.921] (CS), [0.772, 0.924] (biomedicine), [0.846, 0.974] (materials) — wider than link-level Wilson intervals, conclusions unchanged.

From-scratch enumeration (recall side): the same author enumerated terms from zero on 30 further held-out abstracts (10 per domain, frozen test splits; 418 unique terms, exact case-folded surface match). The batch-schema arm covers **0/418** — 0 even among the 83 terms recurring in >10 construction-corpus documents; size-aligned statistical arms cover 16.9–22.9% of that core stratum (C-value 19 > frequency = three-source 14), LLM-direct 47.0% with a 4.6× larger unvalidated lexicon. Half the terms (209) never occur in the 3,000-doc construction corpus — paper coinages that no bounded static lexicon can carry (mention-layer scope), bounding lexicon-layer recall itself. Per 1,000 entries, recall efficiency: C-value 16.6 > three-source 14.8 > frequency 12.2 > LLM-direct 10.5. Three-source is thus a high-validity, bounded-budget *repair* for the entity layer, not a high-recall term extractor; coinages and rare terms are mention-layer scope (the strata table ships with the artifact).

The collapse signature travels (Table 5, Figure 5). The batch-schema arm on CS is the incident in miniature: the domain's *own* committed schema lexicon yields C_{10}=0.899 with blind judged-validity 0.565, 18 matched entities, 5.7% of documents carrying any annotation (its aliases are category-shaped and largely Chinese or uppercase-acronym forms, absent from English abstracts), and silver recall 0.000; a language-aligned control (below) rules out orthographic confounds. Biomedicine and materials schema arms are milder — some of their category words ("disease", "cell", "phase") happen to be usable domain terms there, so their blind validity runs higher (0.688–0.906) while they still concentrate (C_{10} 0.43–0.83) at recall ≤0.009: concentration, not per-link validity, is the collapse signature. Frequency floods at any size: 25–28 concepts per document with validity 0.36–0.45 — volume without terminology. C-value at equal size repairs concentration (0.26–0.55) but not validity (0.33–0.40): in abstract-length corpora, statistical salience is not termhood. Since these baselines' types were originally random, the table reports both under *unified typing* (term sets frozen; types from the same LLM classifier as three-source's classification stage; random-typed rows archived in the long version). Unified typing shifts their validity by −3 to +13 points (frequency 0.405/0.362/0.450, C-value 0.400/0.400/0.333 in the three domains, vs. 0.267–0.400 random)—a bounded confound that nowhere closes the gap to three-source (0.838/0.850/0.912). The dominant deficit is candidate selection: the same classifier accepts only 6–19% of the frequency arm's terms. LLM-direct reaches the highest validity in materials (0.925 [0.846, 0.965]) and the highest silver-P everywhere — but at 4.7–8× the lexicon size with no presence control, and its silver-P is the same-family-inflated column. Among the three *size-matched* arms, three-source holds the best arbitrated validity (model-judged throughout) in CS (0.838 [0.742, 0.903]) and biomedicine (0.850 [0.756, 0.912]) and sits within overlapping intervals of LLM-direct in materials (0.912 [0.830, 0.957] vs. 0.925) and of batch-schema in biomedicine (0.850 vs. 0.906 [0.810, 0.956] — whose recall is 0.009, so its validity lead does not qualify it as a lexicon); its recall is at or above the frequency baseline in two of three domains at one-half the C-value arm's and one-eighth the frequency arm's annotation volume per document. Generic-head share is 0.5% (CS), 3.4% (biomedicine), 7.0% (materials) — reported per domain because the head list is the incident corpus's category vocabulary: its biomedicine/materials drivers ("disease", "drug", "phase", "cell") are legitimate domain terms in those domains, so the cross-domain residual is substantially list–domain mismatch rather than pure genericity (the frequency arm reads 4.5% in CS, where the same words are generic). No single arm dominates all reported metrics; we present three-source as the design point that pairs size-matched validity with bounded annotation volume under this protocol, not as a uniquely optimal balance.

Language confound, resolved by control. Because the CS batch-schema lexicon carries Chinese and uppercase-acronym aliases against English abstracts, its failure could be read as language mismatch rather than semantic shape. A language-aligned control (English-normalized aliases only: CJK dropped, ASCII lowercased, 80 terms) reads C_{10} 0.859 with the same 18.4 entities and the same 5.7% matched documents as the raw arm — the Chinese aliases never matched English text, and the English ones that do are category words. Normalization does move one observable: raw 94.1 links fall to 65.9 (−30%), because case-variant aliases compile to overlapping case-insensitive patterns that each counted the same span; folding duplicates deflates top-head counts unevenly — the whole of the C_{10} shift. A translation control closes the remaining path: the schema's non-ASCII aliases are exactly sixteen type-name translations (the Chinese for "algorithm", "data structure", etc.), not domain-term aliases; translating them to English adds two net-new surfaces with zero corpus hits—every metric identical to the deletion-based alignment. There is no Chinese terminology to lose: the CS collapse is semantic, not orthographic.

What carries the recipe (CS ablation, same frozen samples; Table 6). Removing corpus-candidate classification collapses coverage (concepts/doc 3.60→1.63): classification is the coverage workhorse. Removing presence validation keeps 977 raw entries for almost the same coverage (3.92 vs 3.60) — the gate removes 74% of raw LLM mass at ~8% coverage cost, which is precisely the trade the incident demanded. Seed and expansion alone are thin (1.3–1.5 concepts/doc); unvalidated knowledge seeds carry 13× the entries of validated ones (768 vs 58) for barely more matched entities (80.8 vs 50.9) — mass without coverage. The size-matched frequency control floods (30.7 concepts/doc, 4.5% generic head). The recipe's value is in the assembly and the gate, not in any single source. One component is demoted on this evidence: *expansion*. Removing it costs nothing structurally (entities 211.7 vs. 202.1, concepts/doc 3.73 vs. 3.60), and its standalone hits read 0.433 valid under double judging with only 0.50 judge agreement (n=60) — contested quality. We reframe expansion as an optional diversity module, not a necessary source; the load-bearing pair is corpus-attested classification plus the presence gate.

8.1 Where Recall Is Lost, and What Selection Can Buy

The silver recalls above are occurrence-level against an enumeration instrument; before optimizing selection we decompose where the loss sits. For every silver occurrence we test four nested events: expressible by *some* candidate (*C*: a wide pool — all construct n-grams at frequency ≥2 plus definition-evidenced abbreviations plus all LLM candidate lists; 12,149/15,032/13,121

Table 5: Cross-domain replay: five lexicon arms on identical frozen samples (structural metrics are means over ten disjoint ~690-document samples; |lex| size-matched for freq/cvalue/threesource). Valid = blind judged-valid rate after cross-family arbitration (n=60–80 per cell; Wilson 95% CIs in the text); P/R/F1 are occurrence-level against silver enumerations — levels indicative only, with arm recall orderings judge-stable (see text). Hatched arm in Figure 5: silver-P same-family inflated. †Frequency/C-value rows are the unified-typing re-run (types re-assigned by the same LLM classifier as three-source's classification stage); silver P/R re-run under the unified-typing term sets—random-typed rows are archived in the archival long version and the artifact.

Domain	Arm	Terms	Entities	C_{10}	Conc/doc	Valid	P_s	R_s	F_1
CS	batch schema	95	18.4	**0.899**	1.14	0.565	0.000	0.000	0.000
CS	frequency K=251 †	251	286.0	0.319	24.8	0.405	0.091	0.097	0.094
CS	C-value K=251 †	251	255.0	0.554	5.1	0.400	0.299	0.043	0.075
CS	LLM direct	1198	344.5	0.501	4.6	0.787	0.663	0.132	0.220
CS	three-source	251	202.1	0.404	3.6	**0.838**	0.274	0.050	0.085
Biomed	batch schema	130	100.5	0.428	2.8	0.906	0.089	0.009	0.016
Biomed	frequency K=513 †	513	574.0	0.166	25.1	0.362	0.038	0.038	0.038
Biomed	C-value K=513 †	513	449.0	0.291	6.5	0.400	0.072	0.022	0.034
Biomed	LLM direct	2003	591.5	0.309	4.9	0.838	0.343	0.085	0.136
Biomed	three-source	513	408.7	0.281	5.1	**0.850**	0.273	0.060	0.098
Materials	batch schema	152	58.5	**0.829**	2.3	0.688	0.054	0.004	0.007
Materials	frequency K=384 †	384	448.0	0.151	28.2	0.450	0.028	0.035	0.031
Materials	C-value K=384 †	384	390.0	0.256	6.8	0.333	0.153	0.044	0.068
Materials	LLM direct	2045	713.8	0.198	6.9	**0.925**	0.336	0.093	0.146
Materials	three-source	384	337.4	0.268	5.4	0.912	0.223	0.046	0.076

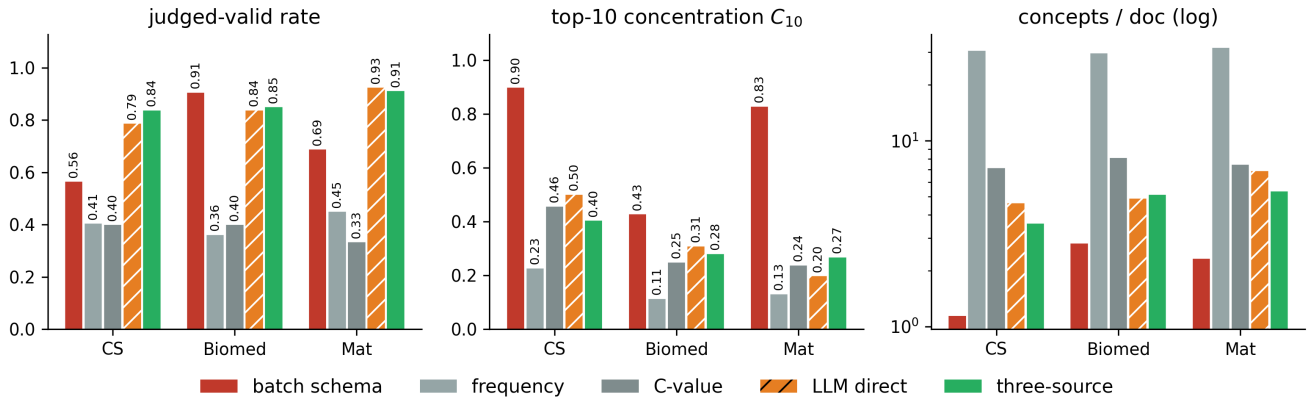


Figure 5: Cross-domain replay at a glance. Left: blind judged-valid rate after cross-family arbitration. Middle: top-10 concentration (the collapse signature; note the schema arm). Right: concepts per document on a log scale (the flooding signature; note the frequency arm). The LLM-direct arm is hatched where its silver-P reads same-family inflated. In the left panel the frequency/C-value bars show the unified-typing re-run (types re-assigned by the shared LLM classifier); random-typed rates archived in the archival long version.

terms per domain), kept in the arm's dictionary (K), detected by the actual scan at that occurrence (D), and linked under normalization (L). The funnel is unambiguous: $P(C)$ is 0.44/0.45/0.41 (CS/biomed/materials) — *the candidate pool, not the ranking, is the binding constraint*; three-source keeps $P(K|C)$=0.15/0.21/0.14; scan and link are nearly lossless ($P(D|K{\cdot}C)$ 0.89–0.98, $P(L)$ ≈1). The pool gaps are two-character element symbols (excluded by the tokenizer's 3-character minimum), terms attested only in test abstracts (construct-side presence cannot see them), and neologisms; the selection gaps include, e.g., *LLM* and *large language model* themselves in CS.

Risk-constrained selection, tested and not adopted. The natural next question is whether a learned or set-selection objective beats per-term ranking at equal budget. On the same pools, with construct-side labels from the cross-family judge (test documents never touch pool, training, or tuning) and proxy link-correctness

Table 6: Source ablation on CS (identical frozen samples; terms column makes the size confound explicit). −presence = all raw components unvalidated.

Arm	Terms	Entities	C_{10}	Conc/doc	Gen-head
full three-source	251	202.1	0.404	3.60	0.005
−seed	221	194.1	0.437	3.35	0.005
−classify	93	70.2	0.601	1.63	0.000
−expand	232	211.7	0.397	3.73	0.004
−presence (raw)	977	260.9	0.374	3.92	0.004
seed-only (valid.)	58	50.9	0.666	1.52	0.000
seed-only (raw)	768	80.8	0.616	1.57	0.000
classify-only	181	169.0	0.463	3.18	0.005
expand-only	42	27.3	0.838	1.29	0.000
freq size-matched	251	312.4	0.227	30.7	0.045

scores (explicitly proxies), neither learned ranking nor coverage-linked selection dominates the three-source pipeline at any tested budget ($K/2$ to $2K$): silver-P rises while arbitrated blind validity collapses in CS and biomedicine (learned-rank 0.29 [0.19, 0.41] and 0.46 [0.32, 0.55] vs. 0.838/0.850) and trails in materials (0.86–0.89 vs. 0.912 at one-fifth the recall); recall is flat in K (the pool ceiling again); and the selected lexicons *concentrate* (C_{10} 0.69–0.73 in CS vs. 0.404) — a proxy-validity objective re-creates the head-heavy distribution our monitors flag, and the silver-P gain was an instrument artifact (enumeration under-cover flatters arms whose extra mass is invalid). We keep the three-source pipeline as the repair baseline and report this as a bounded negative result; construction details, budget curves, and the beam-search reference (within 1.5% of local search) are in the archival long version and the artifact.

9 CONCLUSION

When does a structurally valid lexicon produce semantic collapse? When its entries are category words: the scan executes faithfully, and a dictionary-only swap reproduces the signature. **What does corpus-presence gating repair, and what remains?** Terminology and coverage (annotated documents ∼51.5%→98.5%) — but a generic tail persists; presence is necessary, not sufficient. **What can gold-free diagnostics see?** Distribution and names; per-link truth they cannot see — blind judging shows old-dictionary links are locally as "valid" as new ones (0.94 vs 0.93), so the collapse hides from per-link audits, and the semantic gain runs through vocabulary and mass (equal-size frequency control: 0.45 vs 0.93). All judged rates remain model-assisted with disclosed family exposure; an independent dual-annotator human gold set is the declared next evaluation step. The incident motivated and validated; the controlled swap, the stratified audit, the graded-degradation alarms, and a three-domain public-corpus replay carry the evidence. **Does it generalize?** On three public domains the batch-schema form reproduces the collapse signature (C_{10} up to 0.899 with recall ≤0.009), frequency floods, and the three-source recipe is the size-matched design point with the best arbitrated validity at bounded annotation volume — a trade-off under the tested metrics and AI instrument, not a claimed unique optimum; the failure mode and the repair are properties of the construction method, not of one corpus.

10 LIMITATIONS

(1) The generic tail survives repair (9% of links on "be"). (2) Retrieval attribution is bounded, not eliminated: the full lexicon×configuration decomposition is runnable on frozen slices or isolated stacks via graded degradation — not yet run, rather than impossible. (3) Coverage: two libraries rebuilt, two smaller treated; bilingual anomalies documented, not explained. (4) Six secondary incidents extended the repair; reported as findings. (5) Audit and cross-judge are model-assisted (same vendor family), not human-adjudicated (a 100-item dual-source human annotation sheet ships with the artifact); the risk-stratified sample mixes quota-defined strata and clusters on 187 entities, and unparseable cross-judge responses may be non-random. (6) The isolated experiment uses a semantic-replica scanner (not instrumented against the service path) and committed-snapshot dictionaries over ten resamples of one corpus — a stability check; its metrics are structural, with semantic reading confined to sampled audits (the blind round included: same judge family, term-mention rubric not comparable to the link audit); governance verdicts (removed 0.033 vs retained 0.700) inherit that exposure. (7) The cross-domain study (§8) is model-annotated with a single-author human verification of a 197-term sample (precision bound; recall is referenced by a single-author 30-abstract from-scratch enumeration; §8): silver enumeration agreement across families is 0.41–0.55; an instrument-sensitivity rerun on the shared 30-document subset shows the arms' recall ordering stable under both references (F1 mid-band orderings in CS are instrument-conditional); one gateway judge is unusable for enumeration, and the LLM-direct arm's silver-P is same-family inflated — the cross-domain conclusions rest on structural metrics, recall orderings, and arbitrated blind validity, with the source-substitution adaptation (LLM knowledge seeds in place of seed transplant) disclosed. The recall funnel and the selection study (§8.1) inherit the silver instrument for their loss attribution and training labels; the selection arms' dev labels come from the cross-family judge, and the negative result is stated under those proxies.

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